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30 years of experience in programming (mostly Python and C++) and M.Sc. in software engineering. Specialized in model development for machine learning. Ph.D. in neural network language models. Worked on various vision, language, and speech tasks for 20 years. Developed AI models using PyTorch, TensorFlow, and Theano during the last ten years.

My work has involved writing distributed programs, implementing code for training models, running experiments, and optimizing multi-GPU training. I'm keen to write aesthetic and maintainable code. I want to challenge myself, constantly learn, and stay on top of the research field. I'm motivated by contributing to open source projects and I've also authored several scientific publications.

EDUCATION

Aalto University School of Electrical Engineering

Department of Signal Processing and Acoustics
Espoo, Finland

Doctor of Science (Technology) September 2012 to May 2018

- Research Field: *Speech and Language Technology*
- Thesis: *Modeling Conversational Finnish for Automatic Speech Recognition*

Aalto University School of Science

Department of Information and Computer Science
Espoo, Finland

Licentiate of Science (Technology) September 2007 to September 2012

- Research Field: *Computer and Information Science*
- Thesis: *Finnish Language Speech Recognition for Dental Health Care*

Helsinki University of Technology

Department of Computer Science and Engineering
Espoo, Finland

Master of Science (Technology) September 1999 to August 2006

- Major: *Interactive Digital Media*
- Minor: *Telecommunications Software*
- Extended curriculum in mathematics and physics
- Thesis: *Image-based Detection of Defective Logs*

VISITS

International Computer Science Institute

Berkeley, USA

Visiting Researcher February 2012 to August 2012

- Worked on speech recognition for conversational speech.

Asian Institute of Technology School of Engineering and Technology

Information and Communications Group
Pathumthani, Thailand

Exchange Student August 2005 to December 2005

WORK HISTORY **NestAI**

Helsinki, Finland

Member of Technical Staff

June 2025 to present

- Set up machine learning infrastructure (Google Cloud, LUMI, SLURM, DVC, Apptainer, MLflow).
- Interviewed AI scientists and algorithm developers.
- Planned AI and algorithm development roadmaps.

Verso Vision

Helsinki, Finland

Senior AI Developer

October 2024 to May 2025

- Developed tools for training computer vision models for fall detection (PyTorch, Lightning).
- Developed an embedded face anonymization application for AXIS cameras (AXIS Camera Application Platform).

Groke Technologies

Turku, Finland

Machine Learning Scientist

October 2020 to October 2024

- Developed custom CNN models for the Groke Pro situational awareness system (PyTorch, Lightning).
- Developed augmentations and optimized data pipelines for multi-GPU training on millions of images (LitData, TorchData, Albumentations).
- Experimented with supervised and self-supervised (SimCLR, MoCo) pretraining.
- Developed Kubeflow pipelines for training and testing computer vision models.
- Pretrained and fine-tuned models in the Google Cloud Platform and in SLURM compute clusters.
- Contributed PyTorch code to open source repositories.
- Participated in scientific conferences, researched new methods.

Nuance Communications

Aachen, Germany

Senior Research Scientist

December 2017 to September 2020

NLP/Machine and Deep Learning

- Developed Transformer sequence-to-sequence models for the Nuance DAX report generation system (TensorFlow, PyTorch).
- Ran experiments in multi-GPU SGE and SLURM compute clusters.
- Contributed to public GitHub repositories and published results in conferences.
- Supervised junior scientists.

Aalto University

Espoo, Finland

Doctoral Candidate

January 2011 to November 2017

- Worked on subword, class, and neural network language models.
- Developed [AaltoASR](#) decoder and server backend (C++).
- Developed the [TheanoLM](#) toolkit for training large-vocabulary RNN language models for speech recognition (Theano).
- Collected a conversational Finnish text corpus from the Internet using data selection algorithms.
- Supervised collection of an acoustic training corpus ([DSPCON](#)).
- Experimented with language modeling and automatic speech recognition in SLURM compute clusters.
- Published results in scientific conferences.

Genera Oy

Helsinki, Finland

Software Designer

May 2001 to January 2012

- Implemented new graphical features to display panel software (C++).
- Designed and developed a distributed system for updating content on KONE InfoScreen elevator displays (C++, PHP, JavaScript).
- Developed image analysis algorithms and designed computer vision systems for timber grading and internal quality control (C++).
- Developed Mitla software for timber measurement and refining (Visual Basic).
- Developed configuration script parsers for control and diagnostics panels (Perl).

PUBLICATIONS

Seppo Enarvi, Marilisa Amoia, Miguel Del-Agua Teba, Brian Delaney, Frank Diehl, Guido Gallopyn, Stefan Hahn, Kristina Harris, Liam McGrath, Yue Pan, Joel Pinto, Luca Rubini, Miguel Ruiz, Gagandeep Singh, Fabian Stemmer, Weiyi Sun, Paul Vozila, Thomas Lin, and Ranjani Ramamurthy (2020)

[Generating Medical Reports from Patient-Doctor Conversations using Sequence-to-Sequence Models](#)

In Proceedings of the First Workshop on Natural Language Processing for Medical Conversations

Peter Smit, Siva Reddy Gangireddy, Seppo Enarvi, Sami Virpioja, Mikko Kurimo (2017)

[Character-Based Units for Unlimited Vocabulary Continuous Speech Recognition](#)

In Proceedings of the 2017 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)

Peter Smit, Siva Reddy Gangireddy, Seppo Enarvi, Sami Virpioja, Mikko Kurimo (2017)

[Aalto System for the 2017 Arabic Multi-Genre Broadcast Challenge](#)

In Proceedings of the 2017 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)

Seppo Enarvi, Peter Smit, Sami Virpioja, Mikko Kurimo (2017)

[Automatic Speech Recognition with Very Large Conversational Finnish and Estonian Vocabularies](#)

IEEE/ACM Transactions on Audio, Speech, and Language Processing

Mikko Kurimo, Seppo Enarvi, Ottokar Tilk, Matti Varjokallio, André Mansikkaniemi, and Tanel Alumäe (2017)

[Modeling under-resourced languages for speech recognition](#)

Language Resources and Evaluation (LRE)

Seppo Enarvi, Mikko Kurimo (2016)

[TheanoLM – An Extensible Toolkit for Neural Network Language Modeling](#)

In Proceedings of the 17th Annual Conference of the International Speech Communication Association (INTERSPEECH)

Seppo Enarvi and Mikko Kurimo (2013)

[Studies on Training Text Selection for Conversational Finnish Language Modeling](#)

In Proceedings of the 10th International Workshop on Spoken Language Translation (IWSLT 2013)

Seppo Enarvi and Mikko Kurimo (2013)

[A Novel Discriminative Method for Pruning Pronunciation Dictionary Entries](#)

In Proceedings of the 7th International Conference on Speech Technology and Human-Computer Dialogue (SpED 2013)

SPOKEN LANGUAGES

Finnish (native), English (excellent written and spoken), German (fluent written and fair spoken), Swedish (fair written)

OPEN-SOURCE	TheanoLM
CONTRIBUTIONS	Author of the open source toolkit for language modeling using neural networks.
	AaltoASR
	Contributed to Aalto University speech recognizer.
	Tensor2Tensor
	Contributed to the library of deep learning models from the Google Brain team.
	Fairseq
	Contributed an implementation of the Transformer model with a pointer-generator network to the sequence modeling toolkit from Facebook AI Research.
	PyTorch Lightning Bolts
	Contributed an implementation of various versions of the YOLO object detection model to the repository of PyTorch Lightning models.
	PyTorch Lightning
	Contributed a generic weight averaging callback that supports Stochastic Weight Averaging and Exponential Moving Average to PyTorch Lightning.
LAST UPDATED	February 2026